

The Effects of Cognitive Training on Cognitive Abilities and Everyday Function: A 10-Week Randomized Controlled Trial

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Abstract

With a random groups design, the current study examined whether a 10-week cognitive training could improve healthy older adult's cognitive functions and everyday problem-solving and whether high ecological validity trainings would have greater positive impact upon everyday problem-solving than low ecological validity trainings. Eighty-six healthy Chinese older adult participants were assigned randomly to five groups, including one control group receiving no training and four groups receiving low ecological memory training, high ecological memory training, low ecological reasoning training, and high ecological reasoning training, respectively. Participants were measured pre- and posttraining on spatial working memory, numerical working memory, reasoning, and everyday problem-solving. Results of this study showed that cognitive training significantly improved targeted cognitive functions and everyday problem-solving performance in all the intervention groups. However, high ecological cognitive trainings failed to show superior impact upon everyday problem-solving compared with low ecological cognitive trainings.

Keywords

aging, cognitive training, memory, reasoning, plasticity

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Introduction

Cognitive decline is a well-observed and documented phenomenon in late adulthood (Wolinsky, Weg, Howren, Jones, & Dotson, 2013). At the individual level, it threatens a person's ability to manage daily life tasks and the sense of independence and autonomy. At the society level, it poses great healthcare, social, and economic challenges. As the population in China ages rapidly, the need for optimizing older people's cognitive functions becomes increasingly urgent (Cheng et al., 2012; Li et al., 2016; Zhu, Hu, & Efrid, 2012).

A variety of interventions have been proposed and examined to help maintain and even improve cognitive performance for older adults in the past several decades (Williams & Kemper, 2010). Of them, cognitive training has become popular. It is widely believed that training basic cognitive functions, such as memory and reasoning, can induce neuronal and cognitive plasticity in the aging brain (Chapman & Mudar, 2014; Lee, 2014; Rahe et al., 2015). Of all the existing cognitive training studies, the Advanced Cognitive Training for Independence and Vital Elderly (ACTIVE) may be the most influential one (Ball et al., 2002). Nearly 3,000 healthy older adults were randomly assigned to one of the following four groups: memory training, reasoning training, processing speed training, and no training (i.e., control group). Results of this study showed that training improved participants' cognitive performance on the corresponding functions in all the three experiment groups. Follow-up studies showed that the positive impact of the ACTIVE trainings could last up to 5 years on memory (Willis et al., 2006) and up to 10 years on reasoning and information processing speed (Rebok et al., 2014).

While a growing body of research evidence indicates that cognitive training can help maintain and even improve targeted cognitive functions, the transferability of gains in performance after cognitive training to everyday functions is often questioned (Lustig, Shah, Seidler, & Reuter-Lorenz, 2009; Martin, Clare, Altgassen, Cameron, & Zehnder, 2011). For example, the ACTIVE and its follow-up studies revealed that cognitive enhancement, in general, was observed for the tasks that had been encountered during training (Hertzog, Kramer, Wilson, & Lindenberger, 2009). McDaniel and Bugg (2012) argue that cognitive training is not much practically meaningful if the benefits it produces cannot be generalized to older adults' everyday functions. In addition, everyday functions were rarely measured in previous studies (Kelly et al., 2014). Thus, further work is needed to examine the relationship between cognitive training and everyday functions directly with including everyday functions as dependent variables and, meanwhile, to give considerations on how to enhance training transfer to daily life scenarios when designing cognitive interventions.

As mentioned earlier, previous studies found that cognitive training usually resulted in positive gains on tasks that were similar to the ones learned during training sessions (Hertzog et al., 2009). Accordingly, more and more researchers suggest that training programs should be designed with simulated everyday tasks

to maximize the transfer effect to daily life performance (Cantarella, Borella, Carretti, Kliegel, & de Beni, 2016; Green & Bavelier, 2008; Hering, Rendell, Rose, Schnitspahn, & Kliegel, 2014). Following this suggestion, it is reasonable to expect that cognitive trainings adopting learning tasks at high levels of ecological validity, the degree to which tasks adopted in a controlled experimental condition are similar to those in naturalistic or real world environments (Tupper & Cicerone, 1990), have stronger transfer effect upon everyday functions than those using learning tasks at low level of ecological validity.

The present study was conducted to examine effect of cognitive training on both basic cognitive functions and everyday functions with a sample of healthy older adults in Mainland, China. Training programs of this study were designed to target two basic cognitive functions, including memory and reasoning, with tasks at either high or low levels of ecological validity. We proposed the following research hypotheses: (a) Cognitive trainings would enhance participants' cognitive functions, especially targeted cognitive functions; (b) cognitive trainings would improve participants' everyday functions; and (c) cognitive trainings with high degree of ecological validity would show greater positive impact on daily life functions than those with low degree of ecological validity.

Method

Participants

One hundred and twenty healthy older adults were recruited from the community and screened for eligibility in this study. Inclusion criteria for enrollment in the study included: corrected visual acuity of 20/40 or better in both eyes; normal color vision; hearing loss no greater than 50 dB at 1000 and 2000 Hz; a score of at least 26 on the Mini-Mental Status Exam (Folstein, Folstein, & McHugh, 1975); no evidence of depression; no previous brain surgery, trauma, neurological disorder, or use of antipsychotic or antiepileptic drugs; and living independently. Eighty-six people were eligible for participating in this study, including 50 female and 36 male, with age ranging from 60 to 83 years old (Mean = 68.55, $SD = 5.74$) and education varying from elementary school to college.

Procedure

Participants were randomly assigned to five groups, including one control group and four training intervention groups. Participants of the four intervention groups received a 60-minute cognitive training session each week in small group settings. The entire training intervention program lasted for 10 weeks. The training programs used in the four intervention groups were low ecological (LE) memory training, high ecological (HE) memory training, LE reasoning training, and HE reasoning training, respectively.

LE memory training. Participants were to memorize numbers, words, and instructions. All the memory tasks (e.g., randomly generated number sequences or vocabulary) included in this training were designed with weak connection to older adults' daily life.

HE memory training. Training tasks were designed to simulate activities participants often perform in their daily lives. One sample task is to remember instructions of taking a new prescription drug because older adults often encounter the need to remember instructions to take daily medicines.

LE reasoning training. It included a series of tasks requiring participants to use reasoning based on information available in order to complete a task or to solve a problem. All the exercises involved in this training were distal from participants' daily life scenarios (e.g., what is the next number in the array of 2, 10 18?).

HE reasoning training. It consisted of multiple reasoning tasks that were similar to tasks often performed in participants' everyday life. One sample task is to identify medicines that should not be consumed together with alcohol based on information provided.

A basic mental abilities battery and an everyday problem solving scale, a Chinese version of the Observed Task of Daily Living (OTDL-C; Jiang, 2013), were individually administered to participants before and after the training intervention. Assessment administrators were blinded to participant's group assignment. The basic mental ability battery measured three cognitive functions: spatial working memory, numerical working memory, and reasoning.

Spatial working memory. Each item displayed one pair of 4×4 matrix cards containing a number of Xs, ranging from two to nine. Participants were asked to judge if the distribution patterns of Xs presented on the two cards were identical. Then they were to recall the pattern of Xs displayed on one of the cards. The test consists of 12 items and the possible total score ranges from 0 to 12 points.

Numerical working memory. One-digit addition or subtraction math questions were organized in 12 groups. Number of questions in each group varied from one to nine. For each group, participants were asked to answer all the questions one by one and then recall the answers of all the questions in the sequence of the questions being presented. Same as the spatial working memory test, the possible total score of this test ranges from 0 to 12 points.

Reasoning. The test includes 24 items to assess participants' ability to figure out the pattern in a series of shapes and then to identify the next shape in each series from five candidate shapes. The test time was 3 minutes. Each correct answer scores 1 point and the possible highest score is 24 points.

Everyday functions were measured by the OTDL-C test, which consists of four subscales: taking medicine (13 tasks), using telephone (12), electrical appliances (10), and financial management (10). Score ranges from 0 to 3 for each task on the first three subscales and from 0 to 5 for each task on the financial management measure. So, the possible highest total score of the OTDL-C is 139. The scale's internal consistent reliability measured with alpha coefficient is satisfactorily high: 0.92 for the overall scale, 0.85 for taking medicine, 0.75 for using telephone, 0.82 for electrical appliances, and 0.69 for financial management.

Results

Results of analyses of variance (ANOVAs) upon pretraining measures showed that there was no significant between-group difference on all the dependent variables before training. This indicated that the all the groups had similar baseline scores.

Testing Hypothesis 1: Outcomes in Cognitive Performance

Results of cognitive function measures at both pre- and posttraining as well as difference between pre- and posttraining scores on three scales, including spatial working memory, numerical working memory, and reasoning, were presented in Table 1. To assess impact of cognitive training upon basic cognitive abilities, we performed a repeated-measures (RM) MANOVA on the three cognitive function measures. Results of the RM MANOVA showed a significant interaction between time and training conditions, the Wilk's lambda = 0.578, $F(12, 209) = 4.10$, $p < .001$, indicating significant group differences in changes of cognitive functions from pre- to posttraining. Results of ANOVA with gain scores from pre- to posttraining as independent variables showed significant between-group differences in all the three cognitive functions, including spatial working memory, $F(4,81) = 3.11$, $p < .05$, numerical working memory, $F(4,81) = 3.36$, $p < .05$, reasoning, $F(4,81) = 4.75$, $p < .01$. Post hoc comparisons using the Tukey HSD test indicated: (a) the increase in spatial working memory was greater for participants in the HE memory training group than for those in the control group ($p < .05$), (b) the increase in numerical memory was greater for participants in the LE memory training group than for those in the control group ($p < .01$), and (c) the increase in reasoning was greater for participants in the HE reasoning training group than for those in the control group ($p < .01$). Paired t -test showed that (a) the LE memory training group showed significant gain in numerical working memory, while the HE memory training group showed significant improvement in spatial working memory; and (b) the two reasoning training groups showed significant increase in reasoning performance. Overall, the results support the Hypothesis 1: *Cognitive*

Table 1. Cognitive Abilities: Pre- and Posttraining Scores, and Pre- and Posttraining Difference.

Dependent variable/group	N	Pretest		Posttest		Gain		Pre-post t-test
		Mean	SD	Mean	SD	Mean	SD	
Spatial working memory								
HE memory	19	3.00	1.86	4.11	1.73	1.16	1.79	−2.69**
LE memory	17	3.29	1.49	3.94	1.39	0.65	1.54	−1.73
HE reasoning	15	2.67	1.88	4.13	1.60	1.46	1.51	−3.77**
LE reasoning	15	3.60	2.03	4.33	1.54	0.73	1.87	−1.52
Control	20	3.65	1.60	3.25	1.83	−0.4	1.67	1.07
Numerical working memory								
HE memory	19	4.00	1.56	4.58	1.54	0.58	1.43	−1.77
LE memory	17	3.24	1.79	4.88	1.11	1.64	1.87	−3.63**
HE reasoning	15	3.93	1.22	4.20	0.68	0.27	1.28	−0.81
LE reasoning	15	3.87	1.77	4.13	1.25	0.26	0.96	−1.08
Control	20	3.75	1.59	3.60	1.79	−0.15	1.5	0.45
Reasoning								
HE memory	19	5.00	3.02	7.53	2.86	2.53	2.37	−4.66**
LE memory	17	6.94	3.11	6.82	2.13	−0.12	3.24	0.15
HE reasoning	15	4.53	2.77	7.67	1.88	3.13	2.85	−4.26**
LE reasoning	15	6.07	2.91	7.80	3.42	1.73	2.79	−2.41*
Control	20	5.85	3.07	5.65	3.92	−0.20	3.02	0.30

Note. HE: high ecological; LE: low ecological.

* $p < .05$. ** $p < .01$. *** $p < .001$.

trainings enhance participants' cognitive functions, especially targeted cognitive functions (Figure 1).

Testing Hypothesis 2 and 3: Outcomes in Everyday Problem-Solving Performance

Results of everyday function measures at both pre- and posttraining as well as difference between pre- and posttraining scores were presented in Table 2. We conducted a RM MANOVA to evaluate whether training had impact upon those measured everyday functions, including overall measure, medicine, telephone, electrical appliance, and financial management. There is a significant interaction effect between time and training conditions, the Wilk's $\lambda = 0.597$, $F(16, 239) = 2.75$, $p < .001$, indicating significant group

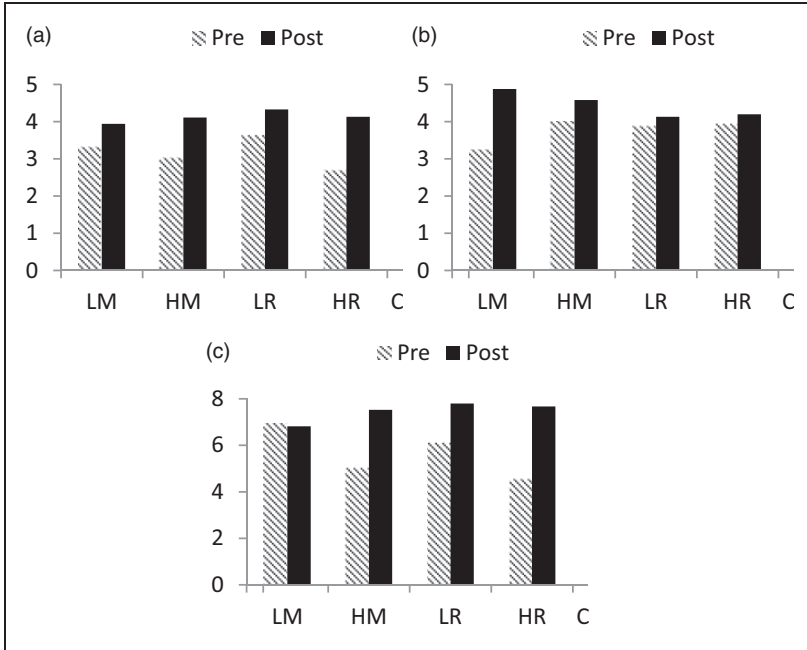


Figure 1. Comparison of three cognitive functions pre- and posttraining: (a) Spatial working memory, (b) Numerical working memory and (c) Reasoning. LM: low ecological memory; HM: high ecological memory; LR: low ecological reasoning; HR: high ecological reasoning.

differences in changes of everyday functions from pre- to posttraining. Results of ANOVA with gain score of overall everyday function measure from pre- to posttraining showed significant between-group differences, $F(4,79)=10.56$, $p<.001$. The follow-up of group comparison analysis showed that gain scores of all the four training groups were significantly higher than the control group but no significant differences among the training groups. Paired t -test showed that (a) participants in all the groups except the control group had significant increase in overall everyday problem-solving performance, telephone, and financial management; (b) the two memory training groups had significant improvement in medicine; and (c) the LE memory and the HE reasoning training groups significantly increased performance in electrical appliances. In conclusion, the results of this study support the research Hypothesis 2 that *cognitive training enhances older adults' everyday functions* but fail to support the research Hypothesis 3 that *compared to cognitive trainings with LE validity, those with HE validity leads to superior improvement in everyday functions*.

Table 2. EverydayFunctions: Pre- and Posttraining Scores, and Pre- and Posttraining Difference.

Dependent variable/group	N	Pretest		Posttest		Gain		Pre-post t-test
		Mean	SD	Mean	SD	Mean	SD	
Overall								
HE memory	19	113.06	21.00	127.32	12.09	14.26	16.00	−3.89**
LE memory	17	123.76	10.98	135.41	7.48	11.65	13.77	−3.49**
HE reasoning	15	117.07	13.46	133.60	8.36	16.53	11.89	−5.39***
LE reasoning	15	122.07	13.43	133.07	11.07	11.00	6.79	−6.27***
Control	20	117.35	24.64	113.00	26.07	−4.35	9.53	2.04
Medicine								
HE memory	19	32.11	7.14	36.68	2.31	4.57	6.83	−2.92**
LE memory	17	34.88	2.85	37.00	3.37	2.12	3.72	−2.35*
HE reasoning	15	35.53	4.70	36.60	2.61	1.07	4.67	−0.89
LE reasoning	15	35.07	3.81	36.87	3.02	1.80	3.71	−1.88
Control	20	33.05	6.40	32.90	7.78	−0.15	4.46	0.15
Telephone								
HE memory	19	27.68	7.35	32.89	4.18	5.21	6.30	−3.43**
LE memory	17	32.06	5.14	35.35	2.40	3.29	6.02	−2.26*
HE reasoning	15	28.33	8.08	34.80	2.54	6.47	8.77	−2.86*
LE reasoning	15	30.07	5.56	34.67	3.48	4.60	4.91	−3.63**
Control	20	30.25	7.97	29.00	9.35	−1.25	5.21	1.07
Electrical appliance								
HE memory	19	25.32	4.87	25.89	4.03	0.58	5.22	−0.05
LE memory	17	25.53	3.47	28.12	2.32	2.59	3.74	−2.85*
HE reasoning	15	25.20	3.90	27.80	2.96	2.60	4.69	−2.15*
LE reasoning	15	25.80	4.25	26.53	4.03	0.73	2.63	−1.08
Control	20	23.30	8.21	21.95	8.51	−1.35	4.34	1.39
Financial management								
HE memory	19	27.95	6.79	31.84	5.15	3.89	5.54	−2.94**
LE memory	17	31.29	4.92	34.94	2.97	3.65	6.23	−2.41*
HE reasoning	15	28.00	6.50	34.40	2.97	6.40	5.69	−4.36**
LE reasoning	15	31.13	5.67	35.00	2.98	3.87	4.36	−3.44**
Control	20	30.75	5.42	29.15	5.52	−1.60	4.04	4.39

Note. HE: high ecological; LE: low ecological.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Discussion

It has been long and widely believed that cognitive training is an effective way to help older adults maintain and enhance cognitive functions and further to improve everyday life performance. Using a sample of healthy older adults in the mainland of China, the current study found that (a) participants' cognitive functions in the four cognitive training groups showed significant improvement after training, especially on the targeted ones, and (b) participants in all the intervention groups performed significantly better on everyday life problem-solving tasks measured by the OTDL-C than those in the control group after completing a 10-week cognitive training. However, the study failed to support the research hypothesis that memory and reasoning trainings with high degree of ecological validity would show greater positive impact on daily life function than memory and reasoning trainings with low degree of ecological validity.

It is worth noting that, although both cognitive functions and everyday problem-solving performance improved after training programs in this study, improvement on everyday problem-solving performance is stronger than gain on cognitive functions. For example, results of ANOVA on posttraining measures only showed that, compared with the control group, the four training groups showed significantly higher scores of overall everyday function performance, $F(4,79)=6.65$, $p < .01$, but no statistically significant differences on all the three cognitive functions, including spatial working memory, $F(4,79)=1.20$, $p > .05$, numerical working memory, $F(4,79)=2.34$, $p > .05$, and reasoning, $F(4,79)=1.64$, $p > .05$. This finding might indicate that basic cognitive abilities, compared with daily life performance, are harder to be improved. Another explanation is that the better performance on everyday problem-solving in the intervention groups relative to the control group could result from the Hawthorne effect which refers to the phenomenon that individuals modify or improve an aspect of their behavior in response to their awareness of being observed (Rapport, Orban, Kofler, & Friedman, 2013).

The current study has several limitations. First of all, each intervention group was trained on only single domain of cognitive functions. Some recent studies (e.g., Cheng et al., 2012) showed that multidomain cognitive training had superior effect compared with single-domain cognitive training. Second, no additional data were collected either between or beyond the pre- and posttraining assessments. Whether participants of those cognitive interventions groups conducted self-training beyond the formal training sessions is unclear. It is reasonable to assume that self-training activities outside the formal training classes could have impact on both immediate and longitudinal effect of training programs.

According to Park and Bischof (2013), most researchers focused on immediate changes in cognitive functions when examining effect of cognitive training interventions and they intentionally or unintentionally ignore the big picture: Many core cognitive abilities, including speed of processing, working memory,

long-term memory, and reasoning, will decline during the normal course of aging regardless of receiving cognitive intervention or not. For example, findings of some studies (e.g., Kliegl, Smith, & Baltes, 1989; Nyberg et al., 2003) suggest that age differences in memory performance were magnified rather than reduced after memory training. Hence, a more conservative view may be that cognitive training can help older adults prevent cognitive abilities from declining or reduce the declining speed. When designing training programs aiming at fighting cognitive decline, more focus can be on executive functions, skills and strategies that help older people handle daily tasks more effectively. For example, it might be more practical to train participants to use written lists in real-life situations rather than to memorize lists using method of loci (Anand et al., 2011). Intensity and time are two important factors affecting outcome of cognitive trainings. A short formal cognitive training program may be able to create long-term mental health benefits if it can change trainees' lifestyle to a more mentally healthy way. Research (e.g., Kuzansky et al., 1998; Lee, 2014; Mireles & Charness, 2002) found that older people's cognitive abilities are better protected if they regularly undertake activities with intensive and complex cognitive requirements such as chess, bridge, and autobiographical writing.

Although the two HE cognitive trainings failed to generate better performance of everyday life problem-solving compared with the other two LE trainings in this study, authors of this article remain to believe that training should be designed with a high degree of ecological validity and relevance to real life scenarios in order to maximize its effect on daily task performance. The two HE cognitive trainings used in the current study may not have been different enough from the two LE cognitive trainings. High-fidelity simulation technique is an option to design better training program. For example, Belchior et al. (2013) used cognitive training with video games to improve older adults' driving skills.

In summary, this study found that cognitive training could improve healthy older adults' cognitive functions and everyday life problem-solving performance. More research is needed to understand what constitutes the most effective type of cognitive training and the underlying mechanism in which cognitive training can help improve daily function, especially in long-term basis.

Declaration of Conflicting Interests

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